THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 9–12: **Light - Reflection and Refraction, The Human Eye and the Colourful World, Electricity, Magnetic Effects of Electric Current** | Class 10 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** straight lines	Rectilinear propagation.
2	**(b)** optical instrument	It forms images like a camera.
3	**(c)** charge	I = Q/t.
4	**(d)** magnetic field	Definition.
5	**(d)** refraction	Definition.
6	**(a)** myopic	Near-sighted.
7	**(a)** its length	R is proportional to length.
8	**(d)** a magnetic field	Oersted's discovery.
9	**(d)** two spherical surfaces	Basic definition.
10	**(c)** two triangular bases and three rectangular lateral surfaces	Glass prism.
11	**(d)** 10 ohm	Add.
12	**(a)** Fleming's left-hand rule	Force direction.
13	**(b)** reflection	Definition.
14	**(b)** cornea	Cornea.
15	**(d)** Rubber	No free electrons.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Light bounces back from a polished surface (1); Surface of the mirror (1).	1+1
17	Retina (2).	2
18	Rate of flow of electric charge (2).	2
19	The region around a magnet where its force is felt (2).	2
20	Convex mirror (1); Virtual and erect (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Shaving: concave (1); Torch: concave (1); Rear-view: convex (1).	1+1+1
22	Iris and pupil control light (1); Cornea and lens focus the light (1); Retina receives the image (1).	1+1+1
23	Q = I t (1); Q = 2 x 5 (1); Q = 10 C (1).	1+1+1
24	Closed curves (1); Outside: N to S (1); Inside: S to N (1).	1+1+1
	OR: n = c/v (1); n = 3×10^8 / 1.25×10^8 (1); n = 2.4 (1).	1+1+1
25	Weakening of ciliary muscles (1); Reduced flexibility of the lens (1); Near point recedes with age (1).	1+1+1
	OR: Length (1); Area of cross-section (1); Nature of the material (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	At infinity: image at F, real, point (1); At C: image at C (1); At C: real, inverted, same size (1); Between P and F: behind the mirror (1); Virtual, erect, enlarged (1).	1+1+1+1+1
	OR: Cornea (1); Iris and pupil (1); Lens and ciliary muscles (1); Retina (1); Optic nerve (1).	1+1+1+1+1
27	Current is proportional to the potential difference at constant temperature (1); V = I R (1); Graph is a straight line through the origin (1); Slope = V / I (1); Slope gives the resistance (1).	1+1+1+1+1

	OR: They emerge from the north pole and end at the south pole (1); They form closed curves (1); The tangent gives the direction of the field (1); Closeness shows strength (1); No two lines intersect (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 9–12: **Light - Reflection and Refraction, The Human Eye and the Colourful World, Electricity, Magnetic Effects of Electric Current** | Class 10 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(d)** tesla	T.
2	**(b)** towards the normal	Water is denser.
3	**(b)** hypermetropic	Far-sighted.
4	**(a)** less resistance	Greater area.
5	**(a)** an electric current	Current deflects the needle.
6	**(b)** centre	Bulges outwards.
7	**(c)** spectrum	Light spectrum.
8	**(a)** 2 ohm	Half.
9	**(d)** the field	Forefinger: field.
10	**(a)** the angle of reflection	Law of reflection.
11	**(b)** iris	Iris.
12	**(d)** ammeter	A for ampere.
13	**(d)** repel each other	N-N repel.
14	**(b)** vacuum	3×10^8 m/s.
15	**(a)** concave lenses	Diverging.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Myopia (near-sightedness) (2).	2
17	Opposition offered to the flow of current (1); R = V / I (1).	1+1
18	Thumb along the current (1); Fingers curl in the direction of the field (1).	1+1
19	Glass (1.5) (1); Water is 1.33 (1).	1+1
20	Most: violet (1); Least: red (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Forefinger: field (1); Middle finger: current (1); Thumb: force (1).	1+1+1
22	1/v - 1/u = 1/f (1); m = v/u (1); m = h'/h (1).	1+1+1
23	No atmosphere in space (1); No scattering of sunlight (1); So the sky appears black (1).	1+1+1
24	1 kW = 1000 W (1); 1 h = 3600 s (1); 1 kWh = 3.6×10^6 J (1).	1+1+1
	OR: Live: red (1); Neutral: black (1); Earth: green (1).	1+1+1
25	Angle of incidence = angle of reflection (1); Incident ray, reflected ray and normal in the same plane (1); Both apply to all reflecting surfaces (1).	1+1+1
	OR: Myopia: far objects blurred; hypermetropia: near objects blurred (1); Myopia image in front of retina; hypermetropia image behind (1); Myopia: concave lens; hypermetropia: convex lens (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	1/v = -1/20 + 1/40 (1); 1/v = -1/40 (1); v = -40 cm (1); Image at C (1); Real, inverted, same size (1).	1+1+1+1+1
	OR: Sunlight falls on raindrops (1); Refraction at the first surface (1); Dispersion into colours (1); Internal reflection (1); Second refraction and colours reach the eye (1).	1+1+1+1+1
27	E = P t = 0.1 kW x 10 h (1); E = 1 kWh (1); I = P / V (1); I = 100 / 220 (1); I = 0.45 A (1).	1+1+1+1+1

	OR: Principle: electromagnetic induction (1); Armature coil, magnets, slip rings, brushes (1); Coil rotates in the field (1); Current is induced and changes direction every half turn (1); Output is alternating current (1).	1+1+1+1+1

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 9–12: **Light - Reflection and Refraction, The Human Eye and the Colourful World, Electricity, Magnetic Effects of Electric Current** | Class 10 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** more resistance	Proportional to length.
2	**(a)** in the opposite direction	The field direction reverses.
3	**(d)** edges	Thin at the centre.
4	**(a)** Violet	Violet.
5	**(a)** stop glowing	The circuit is broken.
6	**(a)** force or motion	Thumb: motion.
7	**(d)** normal	It is the normal.
8	**(a)** pupil	Light enters here.
9	**(d)** voltmeter	V for volt.
10	**(a)** attract each other	N-S attract.
11	**(b)** 3×10^8 m/s	Standard value.
12	**(d)** convex lenses	Converging.
13	**(b)** increases with temperature	Metals.
14	**(d)** concentric circles	Circular.
15	**(d)** optical centre	Optical centre O.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Rs = R1 + R2 + R3 (2).	2
17	Michael Faraday (2).	2
18	Virtual and erect (1); Same size as the object (1).	1+1
19	Iris (2).	2
20	Coulomb (1); About 6.25×10^{18} electrons (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	V = I R (1); V = 0.2 x 10 (1); V = 2 V (1).	1+1+1
22	They are closed curves (1); They do not cross (1); Closer lines mean a stronger field (1).	1+1+1
23	Speed of light changes in water (1); Light slows down in a denser medium (1); It bends towards the normal (1).	1+1+1
24	Splitting of white light into colours (1); Violet deviates the most (1); Red deviates the least (1).	1+1+1
	OR: Rs = R1 + R2 + R3 (1); 3 + 4 + 5 (1); 12 ohm (1).	1+1+1
25	Thumb: direction of motion (1); Forefinger: field (1); Middle finger: induced current (1).	1+1+1
	OR: A straight line through the pole and centre of curvature (1); Normal to the mirror at the pole (1); Also passes through the principal focus (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Image is always erect (1); Image is diminished (1); Field of view is wide (1); Driver sees more traffic (1); Image is virtual and behind the mirror (1).	1+1+1+1+1
	OR: Light enters through cornea and pupil (1); Lens focuses it on the retina (1); Image is real and inverted (1); Retina sends signals through the optic nerve (1); Brain interprets them erect (1).	1+1+1+1+1
27	Length: R increases with length (1); Area: R decreases with area (1); Nature of the material (1); Temperature (1); R = rho L / A (1).	1+1+1+1+1

	OR: A straight wire is connected to a battery and a key (1); A compass is placed near the wire (1); On passing the current the needle is deflected (1); Reverse the current: deflection reverses (1); Current produces a magnetic field (1).	1+1+1+1+1

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